

CDASB: Interactive Version

Conflict-Driven Autonomous System Builder



Abstract

The **Conflict-Driven Autonomous System Builder (CDASB)** is an advanced AI engineering platform designed to automate the end-to-end lifecycle of software development. By moving beyond simple code generation, CDASB implements a "Multi-Agent Dissent" architecture. It ingests complex requirements via natural language or documents (PDF/DOCX), plans architectural solutions, and iteratively builds, tests, and refines them through continuous internal debate. This system ensures that every line of code is challenged and optimized before delivery, providing a robust, human-validated final product.



The Real-World Problem

Most current AI-powered coding tools operate on a "Single-Output" basis, which leads to:

- **Quality Gaps:** AI often generates "hallucinated" or inefficient code that looks correct but fails in production.
- **Lack of Nuance:** Standard prompts struggle to capture deep architectural constraints hidden in large PDF documents.

- **No Self-Correction:** Autonomous systems often proceed with errors without a built-in mechanism for "second thoughts" or critical review.

✓ The Solution

CDASB solves these issues by embedding **Conflict** as a core operation:

- **Structured Debate:** Specialized agents (Critic/Optimizer) specifically look for flaws in the Planner's output.
- **Document Intelligence:** Advanced extraction layers parse PDFs and DOCX files to find hidden requirements.
- **Human-in-the-Loop:** The system presents a transparent reasoning chain and implementation plan for user approval before any code is executed.
- **Iterative Refinement:** A continuous loop of *Code* → *Conflict* → *Test* → *Conflict* → *Improve* ensures maximum stability.

🔧 Technical Architecture

Backend Language **Python 3.12+**

Backend Framework **FastAPI**

Frontend Stack **React / Next.js 15**

Styling **Tailwind CSS**

Database **PostgreSQL**

AI Logic **NVIDIA AI (DeepSeek/Qwen)**

Vector Store **ChromaDB / Pinecone**

Virtualization **Docker Sandbox**

🌐 Languages & Technologies

Languages: Python, TypeScript, JavaScript, SQL, HTML, CSS.

Key Technologies: LlamaIndex (for RAG), LangGraph (Agentic flows), Pydantic (validation), Shadcn/UI (Modern components), PyPDF2/Docx2txt (Extraction).

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